DSA8670 – Week 7 Codex CLI Hands-On

Reflection

Repository Name: LFrager-AmesHousing

I did not have any problems starting the terminal and changing directories to start Codex in the correct folder.

10.13.2026

The initial summarization output I asked Codex to produce for the whole data set was extremely messy. Very hard to read. I should have included instructions to place the data in a table or in a column. It attempted to put the information side by side until it ran out of room and started the next line. Again, it was a very messy output.

This was the first time I used Codex on something this big.

10.14.2026 Started 7:15 pm.

I started with cleaning the data.

I asked Codex: "In the AmesHousing.csv file, remove all rows where the variable Sales Condition != normal and save the file with a new name, ***AmesHousingClean.csv.***" I was totally blown away by how quickly it did it and saved the data in the new data set. It was done in seconds, and it gave me the details on how many rows I now had. I opened the file in Excel and confirmed the other options for sale condition were not there.

So I knew I needed to be specific with the Codex instructions, so each time I included "use the AmesHousingClean.csv data set and told it to write the code using Python". What I don't know is, if I had not included those instructions each time, would it have kept on using Python and would it pull the data from the clean data set or the initial data set? That is something I need to check.

I asked:

Write the code using Python to give me the summary statistics for the following variables: ***Year Built, Gr Liv Area, Bedroom AbvGr, TotRms AbvGr, Garage Area*** using the AmesHousingClean.csv data set." I cut back on the number of features since all of the features were overwhelming.

```
    Year Built  Gr Liv Area  Bedroom AbvGr  TotRms AbvGrd  Garage Area
```

count 2413.000000 2413.000000 2413.000000 2413.000000 2413.000000 mean 1969.436386 1477.410278 2.854538 6.366763 461.254041 std 29.487943 483.061380 0.813430 1.519145 201.675726 min 1872.000000 334.000000 0.000000 2.000000 0.000000 25% 1953.000000 1100.000000 2.000000 5.000000 315.000000 50% 1971.000000 1431.000000 3.000000 6.000000 472.000000 75% 1998.000000 1724.000000 3.000000 7.000000 576.000000 max 2010.000000 4316.000000 6.000000 13.000000 1488.000000

Codex saved over the information it gathered the previous night and gave me the new updated information for the summary. If I would have needed that information that I did on the 13th, I would have lost it, since I did not indicate for Codex to save the code with a new file name.

I asked:

Create the histogram using Python of the SalePrice using the AmesHousingClean.csv data set." Again, it was done in no time and had saved the code file and picture file before I realized with a new file name. I should indicate file names in the process of the ask. It did a good job naming the file correctly - histogram vs graph. I liked that it knew to do that. The picture of the graph was open on my computer screen and I thought I had to save it in the folder. I started doing that and saw it was already there.

This was just a question I wanted to know before setting up the next question to run the code. I asked Codex to figure the median of the SalePrice since I was looking at the question dealing with features and the median SalePrice and it gave me $159,000.

The question I asked:

"I want to know which features predict whether a house sells for above or below the median price using the following features, Year Built, Gr Liv Area, Bedroom AbvGr, TotRms AbvGrd, Garage Area, Garage Type, Kitchen Qual, Overall Qual, Overall Cond. Create the code in Python, using the AmesHousingClean.csv data set. Include figuring the median of the Sale Price with this code."

Since I had asked for the median sale price in a separate question, I figured I needed to include that question in the code. I also used the features that I initially started with and then added a few more to the list. The file it created was the predict_saleprice_class.py. It first gave me the median sale price, how many home above and below the median the confusion matrix, accuracy, precision, recall and f1-score It also gave me the coefficient and odds ratio for these features. The accuracy was 0.894, which is pretty good with the 9 features I selected. I obviously did not select all the features that make the largest difference. We just learned about feature selection in the other class. That would have been good to use to distinguish between the features that are just average vs the features that can be used to make a difference in which home to purchase and renovate.

Below is what I received from the question I asked.

Median SalePrice: $159,000 Target: 1 = Above median; 0 = At or below median Homes above median: 1,196 Homes at or below median: 1,217 Test accuracy: 0.894

Confusion matrix (rows = actual, columns = predicted; 0 then 1): [[225 19] [32 207]]

Classification report: precision recall f1-score support

At or below median 0.88 0.92 0.90 244 Above median 0.92 0.87 0.89 239

accuracy			0.89	483
macro avg	0.90	0.89	0.89	483
weighted avg	0.90	0.89	0.89	483

Model feature effects (sorted by absolute coefficient): - Feature Coefficient Odds ratio - numeric__Gr Liv Area 2.684452 14.650164 - numeric__Overall Qual 1.437387 4.209683 - categorical__Garage Type_2Types -1.229741 0.292368 - numeric__Year Built 1.075751 2.932193 - categorical__Garage Type_Detchd -0.922786 0.397410 - categorical__Garage Type_BuiltIn 0.779614 2.180631 - categorical__Garage Type_Attchd 0.679312 1.972521 - numeric__Garage Area 0.661405 1.937512 - numeric__Overall Cond 0.567953 1.764652 - categorical__Garage Type_CarPort 0.411376 1.508893 - categorical__Kitchen Qual_Ex 0.391656 1.479429 - categorical__Kitchen Qual_Fa -0.391176 0.676261 - numeric__Bedroom AbvGr -0.376240 0.686438 - categorical__Garage Type_Basment 0.339374 1.404068 - categorical__Kitchen Qual_Gd 0.251889 1.286453 - categorical__Kitchen Qual_TA -0.195219 0.822655 - numeric__TotRms AbvGrd 0.024306 1.024603

Coefficient note: numeric features were standardized; categorical features were one-hot encoded.

I also asked:

"Create the code in Python to create the linear regression model using these same variables and data set, with the output SalePrice." Again, I probably should have used all the features to figure out which ones were the best indicators of getting a good house, renovating it and selling it.

Predictors: Year Built, Gr Liv Area, Bedroom AbvGr, TotRms AbvGrd, Garage Area, Overall Qual, Overall Cond, Garage Type, Kitchen Qual Test R-squared: 0.854 Test mean absolute error: $19,340.89 Test root mean squared error: $26,501.81

Intercept: $181,780.49

Model coefficients (sorted by absolute value): Feature Coefficient - categorical___Kitchen Qual_Ex 44653.608329 - numeric___Gr Liv Area 37971.370954 - categorical___Kitchen Qual_TA -20111.996297 - numeric___Overall Qual 16617.839916 - categorical___Kitchen Qual_Gd -16083.128305 - categorical___Garage Type_2Types -15846.879508 - numeric___Year Built 14016.713181 - categorical___Garage Type_Attchd 13318.473394 - numeric___Garage Area 10959.042162 - categorical___Kitchen Qual_Fa -8458.483727 - numeric___Bedroom AbvGr -7922.699417 - numeric___Overall Cond 7187.199278 - categorical___Garage Type_Detchd 3686.290231 - categorical___Garage Type_Basment -2873.060013 - numeric___TotRms AbvGrd -2581.449621 - categorical___Garage Type_BuiltIn 1596.607222 - categorical___Garage Type_CarPort 118.568674

Numeric predictors were standardized, and categorical predictors were one-hot encoded. Coefficients describe model associations, not causation.

I wanted to see the difference in the information it would give me, especially since we have been looking at this kind of information in our other class.

After each ask, I would go and find the file in my folder and run it in the terminal I had open in JupyterLab, reviewing the results as well as the formatting.

If I was to run this again, I would use the Wrapper:Forward Selection approach we just learned about in the other class. I would have seen which features were the best, since it lists the features in order from best to worst for calculating a better AUC each time it did another iteration. Then I could have easily selected the ones it did, or just selected and used the top 10 or 15. I noticed the more I worked with it, before I would click on enter, I would think of something additional to add to the question. The cursor does not move backward easily, so that was a little bit of a challenge.
I had tried typing the question out in Word first, but it did not take the copy and paste option at all.

One thing it did for me. I had used a plural for the column name that I requested, and it corrected it to the column name that was not plural. I made sure that was the correct column.

I tried using ***/transcript*** to get the log of the prompt history from Codex, but that did not work. I had to use ***/export.*** I did this at the end, not at each prompt. I will definitely use it after every prompt. Since I couldn't get / transcript to work, I was afraid I was going to lose the information.

Finished around 9 pm on 10.14.2026.

I tried to get this to run in Windows PowerShell, but it would not. It said Python was not available in this environment. I was able to run the code using Anaconda, JupyterLab and opening a Terminal there.